Homogeneous Systems

Recall from last class:

- Let A be an $m \times n$ matrix, then
$$A = [\vec{a}_1 \ \ldots \ \vec{a}_n], \ \ \vec{a}_j \in \mathbb{R}^m, \ 1 \leq j \leq n.$$
If $x \in \mathbb{R}^n$, then we define
$$A\vec{x} = \vec{a}_1 x_1 + \vec{a}_2 x_2 + \cdots + \vec{a}_n x_n.$$
Thus the matrix equation $A\vec{x} = \vec{b}$ is the same as the vector equation
$$\vec{a}_1 x_1 + \vec{a}_2 x_2 + \cdots + \vec{a}_n x_n = \vec{b}.$$
- $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b} \in \text{Span}(\vec{a}_1, \ldots, \vec{a}_n)$ where $\vec{a}_j$ is the j-th column of A.
- A set of vectors $\vec{v}_1, \ldots, \vec{v}_p$ in $\mathbb{R}^m$ **spans** $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$.
- **Theorem (Theorem 4 in the book)** Let A be an $m \times n$ matrix, then the following are equivalent.
    1. For each $\vec{b} \in \mathbb{R}^m$, the matrix equation $A\vec{x} = \vec{b}$ is consistent.
    2. Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
    3. The columns of A span $\mathbb{R}^m$.
    4. A has a pivot position in every row.
- **Fact (or Theorem 5 from the book):** If A is a $m \times n$ matrix and $\vec{u}, \vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$, then
    1. $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$
    2. $A(c\vec{u}) = c(A\vec{u})$.

Definition 1. *A linear system is called* **homogeneous** *if it can be written as* $A\vec{x} = \vec{0}$ *where* A *is an* $m \times n$ *matrix,* $\vec{x} \in \mathbb{R}^n$*, and* $\vec{0}$ *is the vector of* m *zeros.*

1. Let $A = \begin{bmatrix} 2 & -5 & 8 \\ -2 & -7 & 1 \\ 4 & 2 & 7 \end{bmatrix}$. Find all solutions to $A\vec{x} = \vec{0}$.

Theorem: If $A\vec{x} = \vec{b}$ is consistent for a given $\vec{b}$, and $\vec{p}$ is a solution, then the solution set of $A\vec{x} = \vec{b}$ is the set of vectors of the form $\vec{p} + \vec{v}_h$ where $\vec{v}_h$ is a solution of $A\vec{x} = \vec{0}$.

2. Prove the Theorem.

3. Let $A = \begin{bmatrix} 1 & 0 & 5 \\ -2 & 1 & -6 \\ 0 & 2 & 8 \end{bmatrix}$.

(a) Find all solutions to $A\vec{x} = \vec{0}$.

(b) Let $\vec{b} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$. Describe all solutions to $A\vec{x} = \vec{b}$.

4. Construct a 2 by 2 matrix A such that the solution set to $A\vec{x} = \vec{0}$ is the line in $\mathbb{R}^2$ going through $(0, 0)$ and $(4, 1)$.

Definition 2. *A set of vectors* $\{\vec{v}_1, \ldots, \vec{v}_p\}$ *in* $\mathbb{R}^n$ *is said to be* **linearly independent** *if*

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** *if the equation has non-trivial solutions and the equation is called the* **dependence relation***.*

5. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Is $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ linearly independent?

Let $A = [\vec{a}_1 \ \ldots \ \vec{a}_n]$, then $\{\vec{a}_1 \ \ldots \ \vec{a}_n\}$ are linearly independent if and only if $A\vec{x} = \vec{0}$ has only the trivial solution.

6. Is a set with only one vector always linearly independent?

7. If $\vec{0} \in \{\vec{v}_1, \ldots, \vec{v}_p\}$, then is the set linearly dependent?

8. If $\vec{0} \in \text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$, then the vectors are linearly independent?